

Ioannis Koutalios

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Experience	Sensorfact Data Analyst	<i>Utrecht, Netherlands</i> Mar 2025 – Present
	Leiden University Graduate Student Researcher	<i>Leiden, Netherlands</i> Mar 2022 – Jun 2024
	<i>Teaching Assistant, "Reinforcement Learning" (M.Sc.)</i>	Jan – Jun 2024
	<i>Teaching Assistant, "Science and the Public" (M.Sc.)</i>	Mar 2023 – Jul 2024
	<i>Teaching Assistant, "Astronomy Lab and Observing Project" (B.Sc.)</i>	Mar 2024
Education	Aristotle University of Thessaloniki Undergraduate Student Researcher	<i>Thessaloniki, Greece</i> Sep 2019 – Nov 2020
	<i>Gravitational Waves Group</i>	
	<i>Grading Assistant, "Astronomy and Astrophysics"</i>	Mar 2018 – Jun 2020
	M.Sc., Astronomy and Data Science Leiden University	Sep 2024 <i>Leiden, Netherlands</i>
	B.Sc., Physics Aristotle University of Thessaloniki	Nov 2020 <i>Thessaloniki, Greece</i>
Publications	The resummation model in FLAMINGO: precisely predicting matter power suppression from observed halo baryon fractions van Daalen, M.P., Koutalios, I. , Broxterman, J.C., Wolfs, B.J.H., Helly, J.C., Schaller, M., Schaye, J. <i>Monthly Notices of the Royal Astronomical Society</i> doi: 10.1093/mnras/staf2086	Nov 2025
	Predicting binary neutron star postmerger spectra using artificial neural networks Pesios, D., Koutalios, I. , Kugiumtzis, D., & Stergioulas, N. <i>Physical Review D</i> doi: 10.1103/PhysRevD.110.063008	Sep 2024
	Applying a cross-power model for power suppression in FLAMINGO <i>M.Sc. Thesis, sup. Dr. Marcel van Daalen</i>	Sep 2023 – Jun 2024
Research	Model-Free Reinforcement Learning for Sensorless Adaptive Optics <i>Research Project, sup. Dr. R. Landman, Dr. T. Moerland, Dr. A. Plaat</i>	Oct 2023 – Feb 2024
	Finding the best exoplanets to search for exomoons by radial velocity <i>M.Sc. Thesis, sup. Dr. Matthew Kenworthy</i>	Mar 2022 – Dec 2022
	Spectral Classification of Neutron Star Post-Merger Gravitational Wave Emission <i>B.Sc. Thesis, sup. Dr. Nikolaos Stergioulas</i> <i>Gravitational Waves Group, Aristotle University of Thessaloniki</i>	Sep 2019 – Sep 2020

Conferences	BNAIC/BeNeLearn 2024: Joint International Scientific Conferences on AI	
	Poster: Model-Free RL for Sensorless Adaptive Optics	Utrecht, Nov 2024
	79th Dutch Astronomers' Conference	
	Poster: Applying a cross-power model for power suppression in FLAMINGO	Egmond aan Zee, May 2024
Volunteer	Instructor for the Preparation of the Greek Team for the International Olympiad on Astronomy and Astrophysics	2016 – 2023
	Public Outreach Guide Astronomical Observatory of the Aristotle University of Thessaloniki	2018 – 2021
Distinctions	Scholarship from Cosmote for undergraduate studies	2015 – 2019
	International Olympiad on Astronomy and Astrophysics Honorable Mention (<i>Indonesia</i>)	2015
	Participation (<i>Romania</i>)	2014
Skills	Core Programming & Cloud Python, SQL, AWS, Git (GitHub & GitLab), Linux, $\LaTeX$	
	Data Science & Machine Learning PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy	
	Infrastructure & Tools Prefect, Grafana, Bash, SSH	
Languages	Greek (Native), English (C2), German (B1), Dutch (Beginner)	